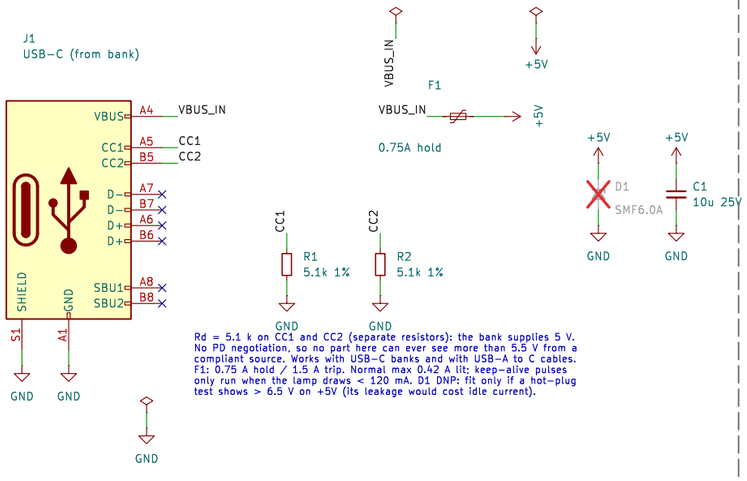
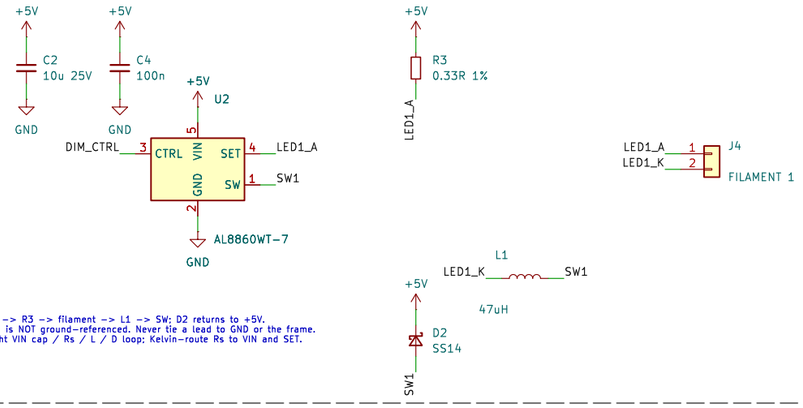
1 USB-C INPUT: 5 V SINK, NO PD



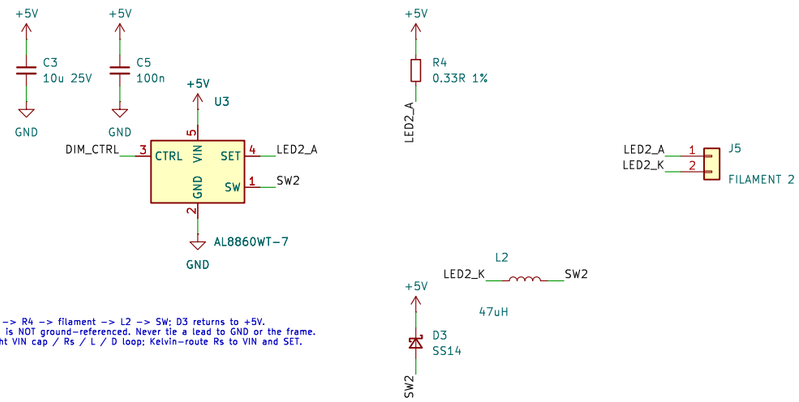
Rd = 5.1 k on CC1 and CC2 (separate resistors): the bank supplies 5 V. No PD negotiation, so no part here can ever see more than 5.5 V from a compliant source. Works with USB-C banks and with USB-A to C cables. F1: 0.75 A hold / 1.5 A trip. Normal max 0.42 A (lit; keep-alive pulses only run when the lamp draws < 120 mA. D1 DNP: fit only if a hot-plug test shows > 6.5 V on +5V (its leakage would cost idle current).

2 FILAMENT 1: AL8860 BUCK, 0.1 V / 0.33 R = 303 mA



Loop: +5V -> R3 -> filament -> L1 -> SW; D2 returns to +5V.
Filament + is NOT ground-referenced. Never tie a lead to GND or the frame.
Layout: tight VIN cap / Rs / L / D loop; Kelvin-route Rs to VIN and SET.

3 FILAMENT 2: AL8860 BUCK, 0.1 V / 0.33 R = 303 mA



Loop: +5V -> R4 -> filament -> L2 -> SW; D3 returns to +5V.
Filament + is NOT ground-referenced. Never tie a lead to GND or the frame.
Layout: tight VIN cap / Rs / L / D loop; Kelvin-route Rs to VIN and SET.

NOTES

Power: one 5 V rail from the bank. Lit 2.06 W (411 mA), idle ~1.1 mA + 3 mA keep-alive (model: power_model.py).

Pins (XIAO):
- D1 GPIO3 ADC_POT ADC1_CH3
- D2 GPIO4 DIM_PWM LEDC 19.5 kHz, 11-bit
- D3 GPIO5 RADAR_OT2 light-sleep wake, high = present
- D4 GPIO6 UART_TX -> radar RX (115200 8N1)
- D5 GPIO7 UART_RX <- radar OT1/TX
- D10 GPIO10 KA_DRV keep-alive gate

Connectors (JST-PH 2.0, through-hole, hand-soldered):
- J2 pot, J3 radar, J4/J5 filaments (pin 1 = +, anode)

Test on the first board (design_review.md, section 6):
- TP3 < 0.2 V with the XIAO held in reset
- each filament 303 mA +/- 5 % at full brightness
- AL8860, L, R12 < 60 C after 45 min in the frame

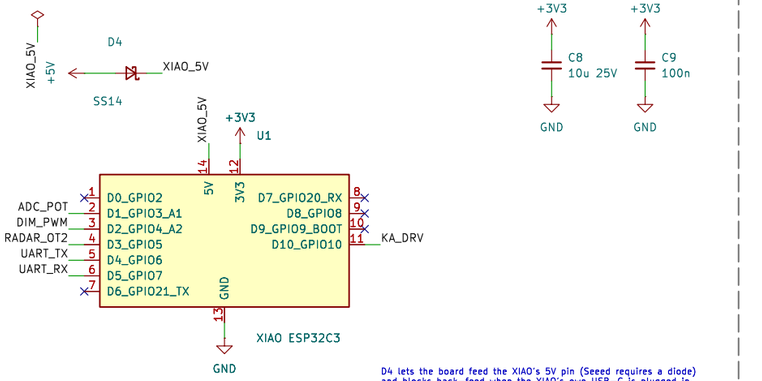
DNP: D1. JLCPCB places everything except U1 and J2-J5.

Test points: TP1 +5V TP2 GND TP3 DIM_CTRL TP4 +3V3 TP5 RADAR_OT2



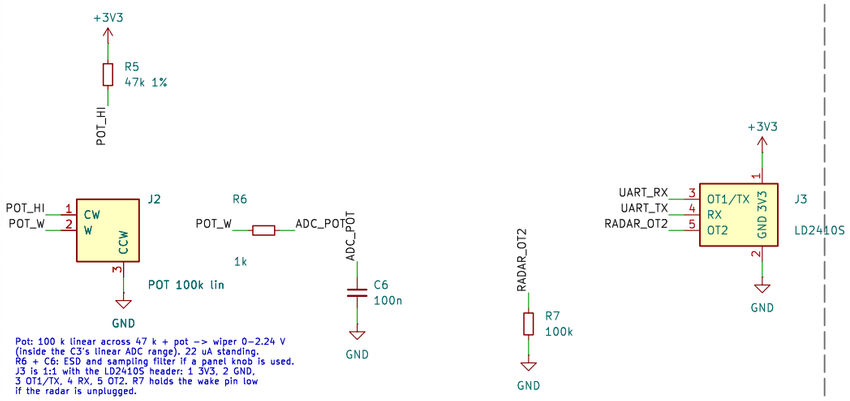
H1-H4: M3, place to suit the frame

4 MCU: SEEED XIAO ESP32C3 (hand-soldered)



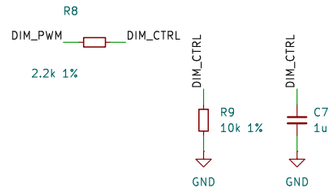
D4 lets the board feed the XIAO's 5V pin (Seeed requires a diode) and blocks back-feed when the XIAO's own USB-C is plugged in. Strapping pins D0/D8/D9 and UART0 D6/D7 are left unconnected. Footprint has only the 14 edge pads; the underside pads (VBAT, EN, JTAG) are omitted so nothing can short to them.

5 BRIGHTNESS POT AND RADAR



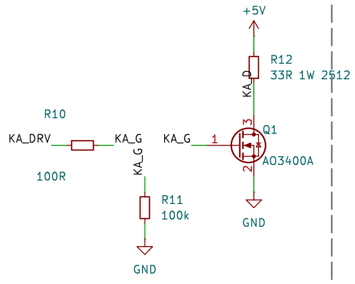
Pot: 100 k linear across 47 k + pot -> wiper 0-2.24 V (inside the C3's linear ADC range). 22 uA standing.
R6 + C6: ESD and sampling filter if a panel knob is used.
J3 is 1:1 with the LD2410S header: 1 3V3, 2 GND, 3 OT1/TX, 4 RX, 5 OT2. R7 holds the wake pin low if the radar is unplugged.

6 DIMMING (both CTRL pins)



Analog (default): 19.5 kHz PWM -> RC -> DC on CTRL.
V_CTRL = 2.70 V x duty. AL8860: 0.3-2.5 V = 0-100 %.
< 0.2 V off, > 2.5 V clamps at 100 %.
Flicker-free: ripple ~19 mV p-p; tau 1.8 ms.
PWM fallback: leave C7 off. PWM <= 500 Hz.
R9 holds CTRL at 0 V while D2 floats (reset, flashing, panic, light sleep) -> filaments dark.

7 BANK KEEP-ALIVE



152 mA for 200 ms every 10 s (3 mA avg) whenever the load is below ~120 mA: dark, or dimmed below ~25 %.
Q1/R10/R11/R12 may be DNP for a bank proven to stay on without it.

Design review: docs/design_review.md
XIAO ESP32C3, HLK-LD2410S radar, brightness pot, power-bank keep-alive
2 x 30 cm 3 V filaments at 303 mA, AL8860 buck, 5 V USB-C input (no PD)

Rob Lim

Sheet: /
File: fgl_g1.kicad_sch

Title: Fused Glass Lighting controller

Size: A3	Date: 2026-10-15	**Rev: G1**
KiCad E.D.A. 9.0.7		Id: 1/1